

Generation Interconnection System Impact Study Report

For Queue Project Q20-1555

Lunenburg 138 kV

April 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1555. The Interconnection Customer proposes a 110.5 MW battery energy storage facility interconnecting at the Lunenburg 138 kV substation in the Virginia area. The request was received on August 21, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 451 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$242,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Helijet to Elk Garden 138kV L: 102.3% loading (282 MVA) following the outage of the Carderock to Elk Garden 138kV; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elk Garden to Helijet 138kV	Reconductor line	\$511,000	47% (\$242,000)
	Total allocated to project		\$242,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.